

Matrix discrepancy and optimization

The Komlós discrepancy conjecture

Difficulty: extreme **Importance:** broadly interesting**Status:** Solved**Provenance:** source-stated conjecture.**Rating rationale:** Dimension-free vector balancing is a foundational discrepancy barrier with wide implications for rounding, optimization and algorithms.

Resolution — 2026-09-11

Affirmative resolution by Shengtao Guo, Ethan X. Fang and Junwei Lu, *Vector Balancing via Directional Total Variation*, [arXiv:2609.11189v1](#), submitted 10 September 2026, **Theorem 1.1, p. 1**. It proves the displayed target with the universal constant $C = 3\sqrt{2\pi}$; its strict bound is stronger than the requested non-strict inequality. Each column of A is one of the theorem’s vectors. All positive dimensions and every column are covered; no algorithmic assertion is required.

Application note by George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology: [proof by reference](#), **Theorem 1** ([PDF](#) · [LaTeX](#)). Stepaniants authors this explanatory note; the discrepancy theorem is credited to Guo, Fang and Lu.

A separate Codex agent checked both the application and the substantive source proof, including its published eigenvalue-convexity input, and returned **PASS: independent review**. This is independent automated-agent verification under the repository’s status definition, not external human peer review or formal certification. The source remains a preprint and discloses Odin AI use. The application draft was prepared with ChatGPT. The original statement and dated history below are retained; the ratings above are historical. [Submission record](#).

Problem statement

Does a finite constant $C > 0$ exist such that, for every pair of positive integers m, n and every matrix $A = (a_{ij}) \in \mathbb{R}^{m \times n}$ satisfying

$$\sum_{i=1}^m a_{ij}^2 \leq 1 \quad (1 \leq j \leq n),$$

there is a vector $x \in \{-1, 1\}^n$ for which

$$\|Ax\|_\infty = \max_{1 \leq i \leq m} \left| \sum_{j=1}^n a_{ij} x_j \right| \leq C$$

The same constant must work for every dimension and matrix. The question asks for existence of a signing; it does not additionally require an algorithm.

This is a matrix balancing problem: choose column signs so that their sum is small in every coordinate. Its norm constraint and simultaneous coordinate control connect discrepancy theory with rounding and linear optimization.

References

1. N. Bansal and H. Jiang, *An Exposition of the $\tilde{O}(\log^{1/4} n)$ Bound for the Komlós Problem*, [arXiv:2608.28452](#) (2026), abstract and introduction, including the explicit conjecture and the new dimension-dependent bound. [Paper](#).
2. N. Bansal and H. Jiang, *Decoupling via Affine Spectral-Independence: Beck-Fiala and Komlós Bounds Beyond Banaszczyk*, [arXiv:2508.03961v2](#) (2025), abstract and introductory results. [Paper](#).
3. W. Banaszczyk, *Balancing vectors and Gaussian measures of n -dimensional convex bodies*, *Random Structures & Algorithms* 12(4) (1998), pp. 351–360, the main vector-balancing theorem underlying the earlier logarithmic bound. [Article](#).

Earlier status check — 2026-09-10

checked the current arXiv records (2608.28452v1, 2026-08-28; 2508.03961v2, 2025-09-09) and searched “Komlos conjecture solved proof September 2026” and “Komlos Bansal Jiang 2026 bound”. The new bound is $O((\log n)^{1/4}(\log \log n)^{7/4})$ in its asymptotic range, not a universal constant. No full proof, counterexample, or withdrawal of these source papers was found. The disproved Hajela conjecture and the separate matrix Spencer conjecture are not this statement.

Audit update (2026-09-10): Rechecked the August 2026 exposition and searched for later Komlós resolutions. Its improved dimension-dependent bound still leaves the universal-constant target open. This is a bounded literature check, not a proof that no solution exists.